Hodge on $X \times X$ Implies Grothendieck's Lefschetz Standard Conjecture B: Lean 4 Formalization

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Abstract

We record a clean, zero-sorry Lean 4 module (StandardConjectures.lean) formalizing the classical implication between the rational Hodge conjecture on the self-product $X \times X$ of a smooth projective variety and Grothendieck's Lefschetz Standard Conjecture $B(X)$. Specifically, let Λ denote the inverse Lefschetz operator L^{-1} on primitive cohomology viewed as a correspondence class on $X \times X$. If Λ is a rational Hodge class and the Hodge conjecture holds on $X \times X$ in the degree of Λ, then Λ is algebraic (LefschetzB Λ). We explicitly separate this implication from the trivial algebraicity of the forward Lefschetz hyperplane cup operator $L = \smile h$ (L_is_algebraic) and document that neither Grothendieck's Conjecture B nor the Clay Hodge Conjecture is unconditionally discharged here.

1 Formalized Theorem Statement

Theorem 1 (hodge_on_product_implies_lefschetz_B)**.** *Let* Λ : Class *represent the Künneth/correspondence class of the inverse Lefschetz operator on* $X \times X$*. Under the hypotheses* hΛ : IsHodge(Λ) *and* hHC : $\forall \gamma$, IsHodge(γ) $\to$ IsAlgebraic(γ) *on* $X \times X$*:*

$$\text{IsHodge}(\Lambda) \ \wedge \ \text{HodgeOnProduct} \implies \text{LefschetzB}(\Lambda).$$

2 Scope and Provenance

- **Origin Commit Timestamp:** Tuesday, September 22, 2026, 18:44:22 CDT (2026-09-22T23:44:22Z).
- **Scope Separation:** $L = \smile h$ is already algebraic (hyperplane section); that is not Standard Conjecture B. Standard Conjectures A, C, and D and unconditional Hodge on $X \times X$ are not claimed.

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A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/StandardConjectures`
- Verified commit:
  `04d0a4aaa8db09ef76219b4a7bd313522b838c7f`
- Lean toolchain: `leanprover/lean4:v4.26.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/StandardConjectures/actions/runs/41568056923`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/StandardConjectures/archive/04d0a4aaa8db09ef76219b4a7bd313522b838c7f.tar.gz`
  `4520f135d01775d090b28bc2619d9e2cf7e6ee9594054fe7b4dce0fcd117eec2`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 2 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`StandardConjectures.hodge_on_product_implies_B_for_all_Hodge`	`(no axioms)`
`StandardConjectures.hodge_on_product_implies_lefschetz_B`	`(no axioms)`